Machine Learning - Short Notes

Chapter 1 What is machine learning

Machine learning is a branch of artificial intelligence that lets computers learn from data instead of following a fixed set of rules written by a developer. The developer gives the program examples and a goal, and the program finds the patterns that reach that goal by itself.

There are three common types. Supervised learning uses labelled examples, such as emails already marked as spam or not spam. Unsupervised learning finds hidden groups in data that has no labels. Reinforcement learning trains an agent with rewards for good actions and penalties for bad ones.

A model starts with random guesses. It predicts, measures the error against the correct answer, and adjusts its internal numbers a little. This loop repeats thousands of times until the error becomes small.

Chapter 2 Documents, chunks and retrieval

Retrieval augmented generation, short RAG, searches your own documents before the language model answers, so the answer can be based on real evidence instead of the model’s memory.

The first step is loading. A document loader reads a file and returns Document objects. Each Document has page_content, the text itself, and metadata, which describes where the text came from.

The second step is splitting. A text splitter cuts the big document into small pieces called chunks, because embedding models work with small pieces of text. Chunk size and chunk overlap control how the cuts are made. Overlap repeats a little text so that a sentence crossing a cut is still understandable in both neighbouring chunks.

Each chunk then becomes an embedding, a list of numbers that represents its meaning, and the embeddings are stored in a vector database. A question is embedded the same way and the closest chunks are retrieved.